CCSU BCA | SEMESTER I

Environmental Science and Sustainability

The course commonly called EWS / EVS

COURSE CODE	BCA-1006T | VAC-I
SEMESTER	BCA Semester I
SYLLABUS	CCSU BCA 2025-26 / NEP

The current BCA course title is Environmental Science and Sustainability. Older CCSU papers use Environmental Studies and a different course code. These notes follow the current four-unit syllabus and use older questions only where topics overlap.

Targeting 80+ is a study goal, not a guaranteed result. Use these notes with class material, the current syllabus and timed paper practice.

How to use this guide

Learn the short unit summary first. Then answer each question without looking, compare your answer with the model, and rewrite it in your own words. For calculations and programs, write every step and test with a small example.

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Unit I | Environment, resources and sustainability

Environment includes interacting physical, biological and human systems. Sustainability asks how people can meet present needs while protecting the ability of future generations to meet theirs.

Core ideas to remember

- The atmosphere, hydrosphere, lithosphere and biosphere interact; human society is part of these systems.
- Natural resources include land, soil, forests, minerals, water and energy.
- Deforestation, mining, dams, erosion, land-use change and over-extraction can affect ecosystems and communities.
- Renewable resources replenish on human timescales only when use remains within regeneration limits; non-renewable resources form much more slowly.
- Intergenerational equity concerns future generations; intragenerational equity concerns fairness among people living today.

Important questions with model answers

Question: Explain sustainability and the two forms of equity.

Model answer

Sustainability means organizing production and consumption so people can meet present needs while maintaining environmental and social conditions for the future. Intergenerational equity asks that current decisions do not unfairly reduce the options of future generations. Intragenerational equity asks whether benefits, costs and access to resources are shared fairly among people today. A strong answer links both ideas to a concrete example such as groundwater use.

Question: Discuss the environmental and social effects of a large dam.

Model answer

A dam can store water, generate electricity and reduce some flood risks. It can also submerge land, alter river flow and sediment movement, affect fish and wetlands, and displace households or tribal communities. Impacts depend on location, design and operation. Planning should assess alternatives, consult affected communities, provide fair resettlement and rehabilitation, protect ecological flows and monitor long-term outcomes.

Unit II | Ecosystems, biodiversity and sustainable practices

An ecosystem is a community of organisms interacting with one another and with non-living surroundings. Biodiversity supports ecological processes and human livelihoods.

Core ideas to remember

- Producers capture energy, consumers feed on organisms and decomposers return nutrients to the environment.
- Energy flows through trophic levels and is lost as heat; matter cycles through ecosystems.
- A food chain is one feeding path; a food web connects many paths. Ecological pyramids summarize number, biomass or energy.
- Threats include habitat loss, overuse, pollution, invasive species and climate change.
- In situ conservation protects species in their natural habitats; ex situ conservation protects them outside, such as in seed banks or zoos.

Important questions with model answers

Question: Describe the structure and function of an ecosystem.

Model answer

The structure includes abiotic factors such as light, water, soil and temperature, and biotic groups such as producers, consumers and decomposers. Producers convert solar energy into chemical energy. Consumers transfer energy by feeding, while decomposers break down dead matter and help return nutrients. Food chains and food webs show feeding links. Ecosystems also provide services such as water regulation, pollination and soil formation.

Question: Compare in situ and ex situ biodiversity conservation.

Model answer

In situ conservation protects species within their natural ecosystems, for example through national parks, wildlife sanctuaries or biosphere reserves. It can conserve habitats and ecological relationships. Ex situ conservation protects species outside their natural habitats, for example in botanical gardens, seed banks or captive breeding programs. It can support recovery when wild populations are critically threatened. The methods can complement one another.

Unit III | Society, legislation and practical applications

This unit connects environmental ethics and development decisions with law, business responsibilities, population trends and climate planning.

Core ideas to remember

- Sustainable development balances environmental protection, social wellbeing and economic activity.
- Businesses can reduce waste and pollution, improve energy and water efficiency, design durable products and report environmental impacts.
- The syllabus names the Water Act 1974, Air Act 1981 and Environment Protection Act 1986; learn their broad purposes and check current amendments in India Code.
- Environmental assessments should consider displacement, consultation, rehabilitation and compensation for project-affected people.
- The National Action Plan on Climate Change includes missions on solar energy, energy efficiency, sustainable habitats, water, Himalayan ecosystems, Green India, sustainable agriculture and strategic climate knowledge.

Important questions with model answers

Question: How can a business contribute to sustainable development?

Model answer

A business can measure its resource use and emissions, prevent pollution at source, use energy and water efficiently, reduce packaging and recover materials for reuse or recycling. It can choose suppliers responsibly, protect worker and community wellbeing and communicate results honestly. These steps are most meaningful when they are built into purchasing, design and operations rather than treated only as public relations.

Question: Name the three environmental laws in the syllabus and state their broad purpose.

Model answer

The Water (Prevention and Control of Pollution) Act, 1974 addresses prevention and control of water pollution. The Air (Prevention and Control of Pollution) Act, 1981 addresses prevention, control and abatement of air pollution. The Environment (Protection) Act, 1986 provides a broader framework for protecting and improving the environment. These are high-level descriptions; amendments and rules should be checked in the official India Code text.

Unit IV | Pollution, waste and sustainable development

Pollution occurs when harmful substances or energy enter the environment at levels that damage health or ecological systems. Prevention is usually more effective than cleanup alone.

Core ideas to remember

- Air, water, soil, noise and marine pollution have different sources and pathways; identify source, exposure, effect and control.
- The greenhouse effect is a natural heat-trapping process intensified by additional greenhouse gases; climate change is the longer-term shift in climate patterns.
- Ozone depletion is mainly associated with ozone-depleting substances in the stratosphere; acid rain forms when certain air pollutants react in the atmosphere.
- Solid-waste management follows prevention, reduction, reuse, recycling, recovery and safe disposal.
- Disaster management includes risk reduction, preparedness, response and recovery; businesses can protect workers, infrastructure and supply chains.

Important questions with model answers

Question: Explain the causes, effects and control of air pollution.

Model answer

Air pollution can come from vehicles, power generation, industry, construction dust and burning of fuels or waste. It can worsen respiratory and cardiovascular illness, reduce visibility, damage crops and buildings, and contribute to regional problems such as acid deposition. Controls include cleaner energy, well-maintained vehicles, public transport, emission standards, dust suppression and avoiding open burning. Monitoring and enforcement help identify whether controls are working.

Question: Differentiate the greenhouse effect, global warming and ozone depletion.

Model answer

The greenhouse effect is the process by which gases in the atmosphere absorb and re-emit heat; it is necessary for a habitable climate. Global warming is the long-term rise in average temperature, strongly increased by human greenhouse-gas emissions. Ozone depletion is the thinning of stratospheric ozone associated with specific chemicals such as chlorofluorocarbons. The three ideas are related to atmospheric chemistry but are not interchangeable.

Paper pattern and topic analysis

The current BCA-1006T course is worth 100 external marks in the 2025-26 syllabus. The archive's 2021-2024 Environmental Studies papers are older shared-course papers and their instructions vary. They cannot establish the current BCA-1006T exam pattern.

The 2021-2024 legacy papers repeatedly include environment components or human-environment relations, natural resources, ecosystems, pollution, biodiversity, acid rain, global warming or ozone depletion, and water conservation. The 2022 and 2023 papers use 60 marks of descriptive answers, 20 of short answers and 20 of very short answers; the 2024 paper has a similar total but different section labels/instructions. These repetitions are useful for overlapping concepts only. The current four units add business sustainability, named legislation and climate missions.

What to prioritize

- Prepare environment and sustainability definitions plus natural-resource examples and impacts.
- Draw a simple food chain or ecosystem diagram; compare conservation methods.
- Learn the three Acts in the syllabus by year and purpose, with current text checked on India Code.
- Distinguish greenhouse effect, global warming, ozone depletion and acid rain.
- Practise 10-mark outlines with definition, causes, effects, controls and a short conclusion.

Last-five-year archive check

For 2021-2024, the archive has BBA/BCA/ETC shared Environmental Studies papers: 2021 (CV-II/CV-III variants), 2022, 2023 and 2024. A 2020 shared Environmental Studies paper is also listed, giving five consecutive legacy years from 2020-2024. No BCA Environmental Science and Sustainability BCA-1006T paper appears in the checked 2025 BCA Semester I listing. These five papers use the older Environmental Studies course and are not presented as five years of the current syllabus.

How to read the evidence

The current four-unit topic map follows the 2025-26 BCA syllabus. The 2022, 2023 and 2024 legacy paper PDFs were readable and were compared for recurring topics and format; the archive lists 2020 and 2021 entries as well. Current-course weighting is not inferred from legacy papers.

Learning resources and source list

The paper archive is a discovery aid, not the university's official syllabus. The current course titles and units below follow the 2025-26 CCSU syllabus reproduced by an affiliated college. Older paper titles are identified as legacy or related material wherever they differ.

CCSU BCA syllabus and unit list (2025-26 reproduction)

CCSUStudy BCA question-paper archive

Environmental Studies paper, December 2022

Environmental Studies paper, December 2023

Environmental Studies paper, December 2024

India Code: Water Act 1974

India Code: Air Act 1981

India Code: Environment Protection Act 1986

United Nations Sustainable Development Goals

Coverage note: no question bank can predict the exact paper. Treat the priority list as a revision map and prepare every listed unit.